ABHILASH KURAPATI

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PROFESSIONAL SUMMARY

AI/ML Engineer with 2+ years of experience building **GCP-native Agentic AI systems**, LLM-powered applications, RAG workflows, and **automated ML pipelines**. Proficient in deploying and monitoring production AI with **Vertex AI**, **Vertex AI Pipelines**, **LangChain**, **Google ADK**, and **MLOps** practices across GKE, BigQuery, Dataflow, Pub/Sub, and Dataproc. Strong background in supervised ML (regression, tree-based, ensemble models), model monitoring, Human-in-the-Loop feedback loops, and CI/CD.

TECHNICAL SKILLS

Programming & Backend: Python, SQL, R, JavaScript, FastAPI, Flask, REST APIs
Generative & Agentic AI: LangChain, LangGraph, Google ADK, RAG Pipelines, Prompt Engineering, MCP Servers, Agentic Workflows
ML Algorithms: Linear/Ridge & Logistic Regression, Random Forest, XGBoost, Time-Series Forecasting, Curve Fitting, Outlier Detection
ML & AI Frameworks: Vertex AI, Scikit-learn, XGBoost, TensorFlow, PyTorch, AutoML
GCP MLOps & Pipelines: Vertex AI Pipelines (Kubeflow), Cloud Composer (Airflow), Model Registry, Cloud Monitoring, GKE, CI/CD
Cloud Data & Storage: BigQuery, Cloud SQL, MongoDB, PostgreSQL, Firestore

EXPERIENCE

Associate AI Engineer | **Pozent Corporation (Johnson & Johnson)**, Piscataway, NJ **Aug 2025 – Present**

- Architected a **GCP-native Agentic AI** system for Medical Impact Assessment reporting, automating PDF and Word generation across structured, semi-structured, and unstructured data sources within secure cloud workflows.
- Built AI workflows on **Vertex AI**, **LangChain**, and **Google ADK**, orchestrating 5+ agentic steps for retrieval, reasoning, content generation, validation, and report assembly at scale.
- Built automated **ML/AI pipelines** with **Vertex AI Pipelines** and **Cloud Composer (Airflow)** orchestrating ingestion (**Pub/Sub**, **Dataflow**), transformation (**Dataproc**, **BigQuery**), and model execution across 2 environments, processing 100,000+ records and 2 **Quickbase** integrations.
- Implemented **GCP-based MLOps**: containerized **FastAPI**/**Flask** inference services on **GKE** via **CI/CD** (Cloud Build/GitHub Actions), with model, prompt, and config versioning for reproducible releases and rollbacks.
- Set up production **model monitoring** with **Cloud Logging/Monitoring** and **BigQuery** dashboards tracking latency, errors, and output quality, with alerts for drift and degradation.
- Integrated **Human-in-the-Loop** review and **RLHF-inspired feedback** loops, using reviewer edits and approvals as labeled feedback to refine prompts and retrieval and drive periodic re-evaluation, cutting report generation from 2–3 weeks to under 1 minute.

Technical Lead | **Open Source with SLU**, Saint Louis, MO **Jan 2025 – Aug 2025**

- Led development of **MeltShiny**, transforming 2 legacy research tools into a cross-platform analytics platform for automated DNA/RNA thermodynamic modeling and reproducible scientific analysis at scale.
- Engineered reactive **R** pipelines with **MeltR** to process melt-curve datasets, apply nonlinear curve fitting, detect outliers, and extract 4 ML-ready thermodynamic features for downstream analytics.
- Implemented **Dockerized** deployment and **GitHub Actions CI/CD**, improving reproducibility across contributors; guided 4 undergraduates through 10+ sprints and reviewed 30+ pull requests.

AI-Full-Stack Intern | **HighRadius**, Remote, India **Jan 2022 – Apr 2022**

- Trained and compared **supervised ML** models (**Linear/Ridge Regression**, **Random Forest**, **Gradient Boosting**) on 50,000+ historical invoices to forecast payment delays and DSO trends for collections.
- Engineered 15+ features and tuned **scikit-learn** models via **GridSearchCV** and cross-validation, tracking **MAE**/**RMSE** vs. baselines.
- Deployed the model as a **Python/Flask** prediction service for the Spring Boot backend, with periodic retraining on new data.
- Containerized **Spring Boot**/**MySQL** with **Docker** (98% **CI/CD** success); tested with **PyTest** (80%+ coverage).

PROJECTS

End-to-End ML Pipeline on GCP | Vertex AI Pipelines, BigQuery, XGBoost, FastAPI **Personal Project – 2026**

- Built an automated **Vertex AI Pipelines** workflow (ingest, feature engineering, train, evaluate, register, deploy) over **BigQuery** data, comparing **Logistic Regression**, **Random Forest**, and **XGBoost**.
- Served the best model via **FastAPI** on Vertex endpoint/GKE with drift monitoring and a retraining trigger that reruns the pipeline when metrics or drift cross a threshold.

Agentic Market Researcher | Claude API, Prompt Caching, Multi-Agent Reasoning, FastAPI **Personal Project – 2026**

- Launched an **agentic AI** market research platform that synthesized 20+ news, SEC/EDGAR, and Reddit sources through asynchronous Python ETL pipelines and structured caching.
- Orchestrated **Claude Haiku** and **Claude Sonnet** agents with prompt caching and weighted ranking, replacing 3+ hours of manual research with one-click insights.

AI SQL Data Retrieval Agent | LangChain, OpenAI, PostgreSQL, Prompt Engineering, React **Graded Project – 2025**

- Built an **LLM-powered SQL agent** using 3-stage **LangChain** workflows (generate, execute, format) for 5+ query types across 10+ PostgreSQL tables, with **FastAPI** and **React**.

EDUCATION

Saint Louis University **Aug 2023 – May 2025**
Master of Science in Computer Science, CGPA: 3.94 / 4.0 Missouri, United States

Chandigarh University **Aug 2019 – Jun 2023**
Bachelor of Engineering in Electronics & Communication Engineering, CGPA: 8.11 / 10 Punjab, India